

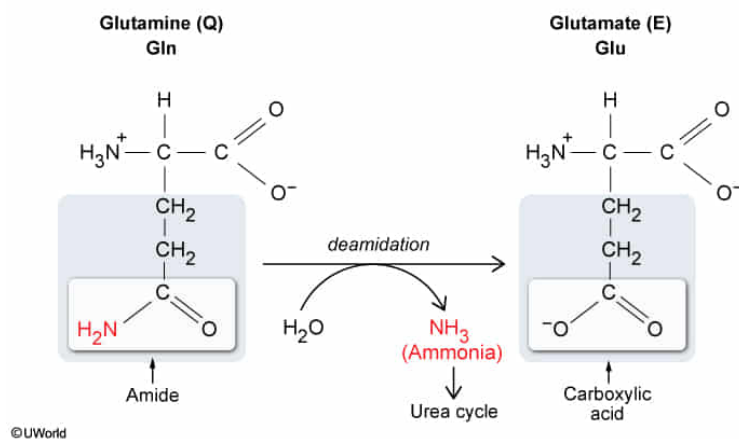
Amino acid catabolism releases nitrogen in the form of ammonia. In the liver, the urea cycle prepares ammonia for excretion. Which amino acid could undergo deamidation to produce ammonia for the urea cycle?

- ☐ A. Lysine [23%]
- ☐ B. Arginine [27%]
- ✓ ☒ C. Glutamine [42%]
- ☐ D. Aspartate [7%]

Correct

41% answered correctly

Explanation:



Amino acids all have a common **backbone structure** that consists of an amino group bonded to a carbon atom (the α -carbon), which is bonded to a carboxylic acid. The α -carbon is also bonded to a side chain, called an R-group, which gives the amino acid its identity.

Amino acid catabolism (degradation) involves the removal of nitrogen from backbones and certain R-groups in the form of **ammonia**, which is detoxified and prepared for excretion by the **urea cycle**:

1. The nitrogen of amino acid **backbones** is removed by **transamination and deamination** (removal of an amine to

produce a ketone), producing α -keto acids in the process.

2. The side chains of **asparagine** and **glutamine** contain **amides**, which are functional groups consisting of a nitrogen atom linked to a carbonyl carbon. Amides are catabolized by **deamidation** rather than **deamination**. Deamidation releases ammonia from an **amide instead of an amine**; the result of deamidation (eg, of Asn or Gln side chains) is replacement of the amide with a carboxylic acid, whereas the result of deamination (eg, of the backbone α -amine of any amino acid) is replacement of the amine with a carbonyl oxygen.

Among the answer choices, only **glutamine** contains an amide group and can therefore produce ammonia through deamidation.

(Choice A) **Lysine** contains an **amine group** in its side chain, not an amide. As such, it can undergo **deamination** but not **deamidation**.

(Choice B) **Arginine** contains a **guanidinium group** in its side chain. This group *does not contain* a carbonyl, so it cannot be an amide group. (Note: Do not confuse arginine with **asparagine**, which *does* have an amide in its side chain.)

(Choice D) **Aspartate** is the deamidated form of asparagine. It contains a carboxylate group in its side chain, not an amide.

Educational objective:

The urea cycle processes ammonia for excretion in the urine. The ammonia that enters the cycle is derived from the amino groups of amino acid backbones (deamination) and from the amide side chains of glutamine and asparagine (deamidation).

Time spent:0 sec

Subject:
Biochemistry

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System:
Amino Acids and Proteins